PART A — QUESTIONS

AF6-001
Question:
Solve: $8x - 7 = 49$
A) 5
B) 6
C) 7
D) 8

AF6-002
Question:
If $6n + 4 = 40$, what is n?
A) 5
B) 6
C) 7
D) 8

AF6-003
Question:
Simplify: $3(5x - 4) + 2x$
A) $15x - 4$
B) $17x - 4$
C) $15x - 12$
D) $17x - 12$

AF6-004
Question:
A repair service charges $30 plus $11 for each hour of work. Which expression represents the total charge for h hours?
A) $30h + 11$
B) $11h + 30$
C) $41h$
D) $30 - 11h$

AF6-005
Question:
If $f(x) = 2x + 13$, what is $f(8)$?
A) 16
B) 21
C) 29
D) 34

AF6-006
Question:
Which ordered pair satisfies the equation $y = 6x - 1$?
A) (1, 6)
B) (2, 11)
C) (3, 16)
D) (4, 21)

AF6-007
Question:
Solve: $7(x - 2) = 35$
A) 5
B) 6
C) 7
D) 8

AF6-008
Question:
A pattern begins: 4, 10, 16, 22, ... What is the next number?
A) 26
B) 27
C) 28
D) 30

AF6-009
Question:
Solve: $x/5 = 11$
A) 16
B) 45
C) 55
D) 60

AF6-010
Question:
A line has a slope of -4 and crosses the y-axis at 9. Which equation represents the line?
A) $y = 9x - 4$
B) $y = -4x + 9$
C) $y = 4x + 9$
D) $y = -9x + 4$

AF6-011
Question:
If 5 boxes contain 65 washers, how many washers are in 9 boxes at the same rate?
A) 104
B) 117
C) 125
D) 130

AF6-012
Question:
Solve: $24 - 3x = 9$
A) 3
B) 5
C) 6
D) 11

AF6-013
Question:
Which expression is equivalent to $18b - 7b - 5$?

A) 11b - 5
B) 25b - 5
C) 11b + 5
D) 18b - 12

AF6-014
Question:
A worker has 22 forms completed and completes 9 more forms each hour. Which equation gives the total number of forms F after h hours?
A) F = 22h + 9
B) F = 9h + 22
C) F = 31h
D) F = 22 - 9h

AF6-015
Question:
If y = -5x + 18, what is y when x = 3?
A) -3
B) 3
C) 15
D) 33

AF6-016
Question:
Solve: 11x + 5 = 6x + 35
A) 5
B) 6
C) 7
D) 8

AF6-017
Question:
Which value of x makes 5x - 6 greater than 24?
A) 5
B) 6
C) 7
D) 4

AF6-018
Question:
A table shows a linear relationship.

x: 0, 1, 2, 3
y: 4, 10, 16, 22

What is the rule?
A) y = 4x + 6
B) y = 6x + 4
C) y = 10x - 6
D) y = x + 4

AF6-019
Question:
Solve: $4(x + 5) = 2x + 34$
A) 5
B) 6
C) 7
D) 8

AF6-020
Question:
If $u = -3$ and $v = 8$, what is $2u + v$?
A) -14
B) -2
C) 2
D) 14

AF6-021
Question:
Which point has a positive x-coordinate and a negative y-coordinate?
A) (-6, 2)
B) (6, -2)
C) (-6, -2)
D) (6, 2)

AF6-022
Question:
A crew assembles 45 units in 5 hours. At the same rate, how many units will the crew assemble in 8 hours?
A) 64
B) 72
C) 80
D) 90

AF6-023
Question:
Solve for q: $V = 6q + 18$
A) $q = V - 18$
B) $q = (V - 18)/6$
C) $q = 6(V - 18)$
D) $q = V/6 + 18$

AF6-024
Question:
What is the slope of a line that goes through (3, 4) and (7, 16)?
A) 2
B) 3
C) 4
D) 12

AF6-025
Question:

Simplify: $13x + 5 - 6x - 12$
A) $7x - 7$
B) $19x - 7$
C) $7x + 17$
D) $19x + 17$

AF6-026
Question:
If $g(x) = x^2 + 5x$, what is $g(5)$?
A) 25
B) 30
C) 45
D) 50

AF6-027
Question:
A sequence follows the rule “add 3, then multiply by 4.” If the first term is 2, what is the second term?
A) 8
B) 12
C) 20
D) 24

AF6-028
Question:
Solve: $6(x - 1) + 3 = 39$
A) 5
B) 6
C) 7
D) 8

AF6-029
Question:
A line crosses the y-axis at -4 and falls 5 units for every 1 unit to the right. Which equation represents the line?
A) $y = 5x - 4$
B) $y = -5x - 4$
C) $y = -4x - 5$
D) $y = 4x - 5$

AF6-030
Question:
If $8/11 = x/33$, what is x?
A) 16
B) 22
C) 24
D) 27

AF6-031
Question:
Solve: $-8x + 12 = -52$

A) -8
B) -5
C) 5
D) 8

AF6-032
Question:
A container starts with 320 fasteners. Each job uses 20 fasteners. Which expression gives the number of fasteners left after j jobs?
A) 320 + 20j
B) 320 - 20j
C) 20j - 320
D) 320 ÷ 20j

AF6-033
Question:
Which equation has x = 12 as its solution?
A) 2x + 5 = 27
B) 3x - 6 = 30
C) x/3 + 4 = 9
D) 5x - 8 = 50

PART B — ANSWER KEY + EXPLANATIONS

AF6-001 — Correct: C — Explanation:
Solve 8x - 7 = 49. Add 7 to both sides: 8x = 56. Divide by 8: x = 7. The most common mistake is subtracting 7 instead of adding 7.

AF6-002 — Correct: B — Explanation:
Start with 6n + 4 = 40. Subtract 4 from both sides: 6n = 36. Divide by 6: n = 6. A common mistake is stopping at 36 and forgetting to divide by 6.

AF6-003 — Correct: D — Explanation:
Distribute first: 3(5x - 4) = 15x - 12. Then add 2x: 15x + 2x - 12 = 17x - 12. A common mistake is forgetting to multiply -4 by 3.

AF6-004 — Correct: B — Explanation:
The fixed charge is $30. The hourly charge is $11 for each hour, so that part is 11h. The total charge is 11h + 30. A common mistake is making 30 the hourly rate.

AF6-005 — Correct: C — Explanation:
Substitute 8 for x: f(8) = 2(8) + 13 = 16 + 13 = 29. A common mistake is multiplying correctly but forgetting to add 13.

AF6-006 — Correct: B — Explanation:
Test the choices in y = 6x - 1. For (2, 11), 6(2) - 1 = 12 - 1 = 11, which matches the y-value. A common mistake is choosing a point that is close but does not make the equation true.

AF6-007 — Correct: C — Explanation:
Solve 7(x - 2) = 35. Divide both sides by 7: x - 2 = 5. Add 2: x = 7. A common mistake is adding 2 before undoing the multiplication by 7.

AF6-008 — Correct: C — Explanation:
The pattern increases by 6 each time: 4, 10, 16, 22. Add 6 to 22: 28. A common mistake is assuming the next number is 27 because the pattern looks close to adding 5.

AF6-009 — Correct: C — Explanation:
$x/5 = 11$ means x divided by 5 equals 11. Multiply both sides by 5: $x = 55$. A common mistake is adding 5 and 11 instead of multiplying.

AF6-010 — Correct: B — Explanation:
Slope-intercept form is $y = mx + b$. The slope is -4 and the y-intercept is 9, so the equation is $y = -4x + 9$. A common mistake is switching the slope and y-intercept.

AF6-011 — Correct: B — Explanation:
Find the unit rate: 65 washers ÷ 5 boxes = 13 washers per box. For 9 boxes: $13 \times 9 = 117$. A common mistake is adding 4 boxes and 4 washers instead of using the rate.

AF6-012 — Correct: B — Explanation:
Solve $24 - 3x = 9$. Subtract 24 from both sides: $-3x = -15$. Divide by -3: $x = 5$. A common mistake is losing the negative sign after subtracting 24.

AF6-013 — Correct: A — Explanation:
Combine like terms: $18b - 7b = 11b$. The constant -5 stays the same. The expression becomes $11b - 5$. A common mistake is combining the constant with the variable terms.

AF6-014 — Correct: B — Explanation:
The worker starts with 22 completed forms. Then 9 more forms are completed each hour. After h hours, the total is $F = 9h + 22$. A common mistake is making 22 the hourly rate.

AF6-015 — Correct: B — Explanation:
Substitute $x = 3$ into $y = -5x + 18$: $y = -5(3) + 18 = -15 + 18 = 3$. A common mistake is treating $-5(3)$ as positive 15.

AF6-016 — Correct: B — Explanation:
Solve $11x + 5 = 6x + 35$. Subtract 6x: $5x + 5 = 35$. Subtract 5: $5x = 30$. Divide by 5: $x = 6$. A common mistake is subtracting 5 from only one side.

AF6-017 — Correct: C — Explanation:
Solve $5x - 6 > 24$. Add 6: $5x > 30$. Divide by 5: $x > 6$. The only listed value greater than 6 is 7. A common mistake is choosing 6, but $5(6) - 6 = 24$, not greater than 24.

AF6-018 — Correct: B — Explanation:
The y-values increase by 6 each time x increases by 1, so the slope is 6. When $x = 0$, $y = 4$, so the y-intercept is 4. The rule is $y = 6x + 4$. A common mistake is using 4 as the slope because it is the first y-value.

AF6-019 — Correct: C — Explanation:
Distribute: $4x + 20 = 2x + 34$. Subtract 2x: $2x + 20 = 34$. Subtract 20: $2x = 14$. Divide by 2: $x = 7$. A common mistake is forgetting to multiply 5 by 4.

AF6-020 — Correct: C — Explanation:
Substitute $u = -3$ and $v = 8$: $2u + v = 2(-3) + 8 = -6 + 8 = 2$. A common mistake is treating $2(-3)$ as positive 6.

AF6-021 — Correct: B — Explanation:
A positive x-coordinate and a negative y-coordinate means the point is in Quadrant IV. The point $(6, -2)$ fits this description. A common mistake is choosing $(-6, 2)$, which has the signs reversed.

AF6-022 — Correct: B — Explanation:
45 units in 5 hours means $45 \div 5 = 9$ units per hour. In 8 hours: $9 \times 8 = 72$. A common mistake is multiplying 45 by 8 without finding the hourly rate first.

AF6-023 — Correct: B — Explanation:
Start with $V = 6q + 18$. Subtract 18 from both sides: $V - 18 = 6q$. Divide by 6: $q = (V - 18)/6$. A common mistake is dividing V by 6 first and then subtracting 18.

AF6-024 — Correct: B — Explanation:
Slope equals change in y divided by change in x. From $(3, 4)$ to $(7, 16)$, y increases by 12 and x increases by 4. The slope is $12 \div 4 = 3$. A common mistake is using 12 as the slope without dividing by the change in x.

AF6-025 — Correct: A — Explanation:
Combine like terms: $13x - 6x = 7x$, and $5 - 12 = -7$. The simplified expression is $7x - 7$. A common mistake is writing $7x + 7$ because the negative sign before 12 is missed.

AF6-026 — Correct: D — Explanation:
Substitute 5 for x: $g(5) = 5^2 + 5(5) = 25 + 25 = 50$. A common mistake is calculating 5^2 as 10 instead of 25.

AF6-027 — Correct: C — Explanation:
Start with 2. Add 3: 5. Then multiply by 4: 20. A common mistake is multiplying first and then adding 3.

AF6-028 — Correct: C — Explanation:
Solve $6(x - 1) + 3 = 39$. Distribute: $6x - 6 + 3 = 39$. Combine: $6x - 3 = 39$. Add 3: $6x = 42$. Divide by 6: $x = 7$. A common mistake is forgetting to combine -6 and +3.

AF6-029 — Correct: B — Explanation:
"Falls 5 units for every 1 unit to the right" means the slope is -5. The line crosses the y-axis at -4, so the equation is $y = -5x - 4$. A common mistake is using a positive slope because the number 5 is given.

AF6-030 — Correct: C — Explanation:
Use the proportion $8/11 = x/33$. Since $11 \times 3 = 33$, multiply 8 by 3: $x = 24$. A common mistake is adding 22 to the numerator because 11 becomes 33.

AF6-031 — Correct: D — Explanation:
Solve $-8x + 12 = -52$. Subtract 12 from both sides: $-8x = -64$. Divide by -8: $x = 8$. A common mistake is giving -8 because both sides contain negative numbers.

AF6-032 — Correct: B — Explanation:
The container starts with 320 fasteners. Each job uses 20 fasteners, so j jobs use 20j fasteners. The number left is 320 - 20j. A common mistake is adding 20j even though fasteners are being used.

AF6-033 — Correct: B — Explanation:
Test x = 12. In choice B, 3(12) - 6 = 36 - 6 = 30, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.